

Answers to questions of  
Grünbaum and Loewner.  
(joint with S. Myroshnychenko  
and K. Tatarko).

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$K$  convex body in  $\mathbb{R}^n$ .

Centroid of  $K$  is

$$C(K) = \frac{1}{|K|} \int_K x \, dx$$

$\downarrow$   
 $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$



Q (Grünbaum). Is  $C(K)$  the centroid of at least  $n+1$  hyperplane sections through  $C(K)$ ?

Q (Loewner)  
Let  $\mu(K)$  be the number of hyperplane sections through  $C(K)$ , whose centroid is the same as  $C(K)$ .

$$\mu(n) = \min_{\substack{K \subset \mathbb{R}^n \\ \text{convex bodies}}} \mu(K)$$

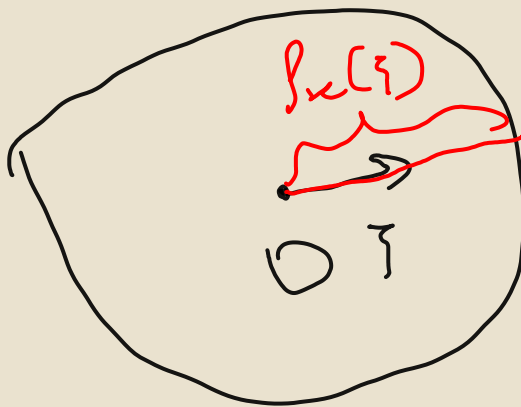
What is the value of  $\mu(n)$ ?

$$\underline{n=2} \quad \mu(2) = 3 = \frac{n+1}{2^n - 1}$$

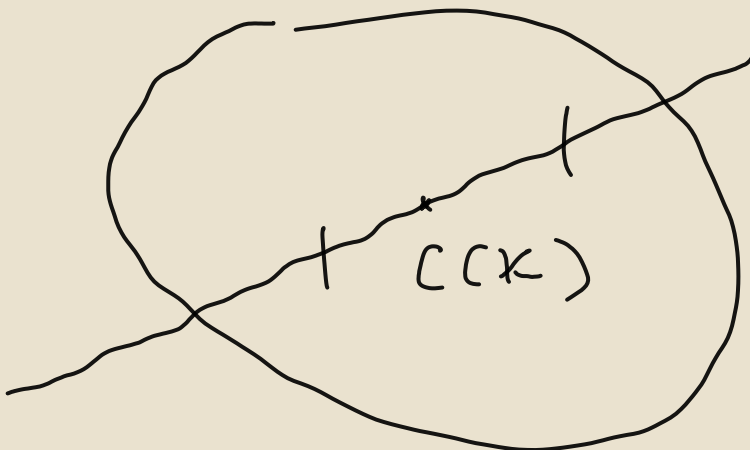
Radial function.

$$\underline{0 \in \text{int } K}$$

$$\rho_K(\xi) = \max_{\xi \in S^{n-1}} \{ \lambda > 0 : \lambda \xi \in K \}$$



$$\underline{C(K) = 0}$$

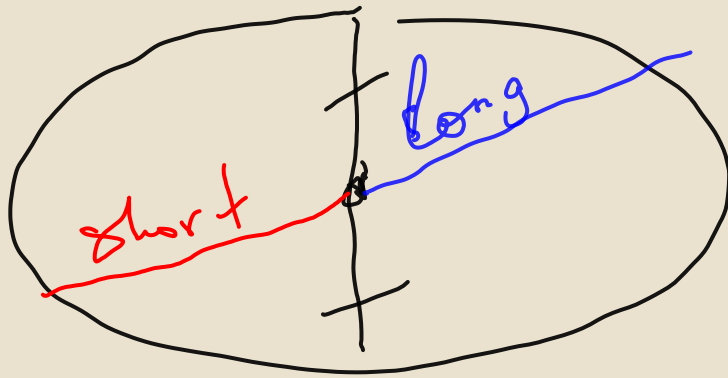


$$\rho_K(\xi) - \rho_K(-\xi)$$

odd, \$C(S')\$

$$\exists \xi : f_k(\xi) = f_k(-\xi).$$

$\nearrow$  Assume that such  $\xi$  is unique. ( $\xi = e_2$ )



$$0 = \int_K x_1 dx = \int_{S'} \int_0^{x=r\theta} r^2 \theta, dr d\theta$$

$$= \frac{1}{3} \int_{S'} \int_x^3 f_x(\theta) \theta, d\theta$$

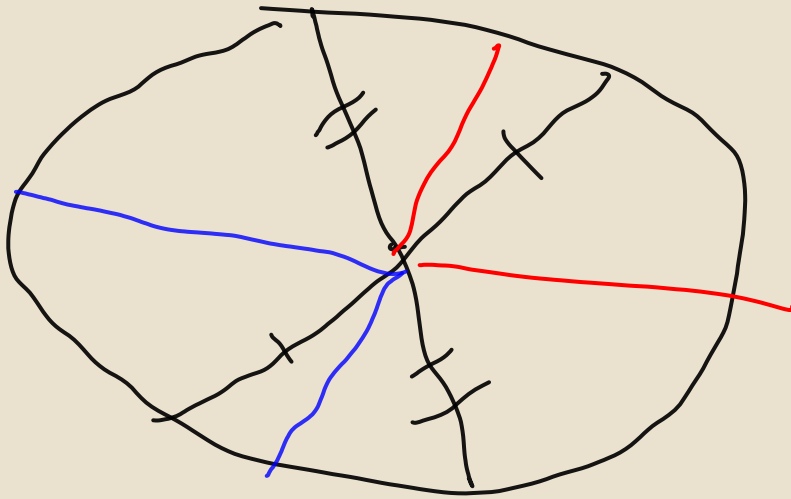
$$= \frac{1}{3} \left( \int_{S' \cap \{\theta > 0\}} \int_x^3 f_x(\theta) \theta, d\theta + \int_{S' \cap \{\theta < 0\}} \int_x^3 f_x(\theta) \theta, d\theta \right) = 0$$

$$\theta \rightarrow -\theta$$

$$\int_{S^1 \cap \{ \theta, \theta > 0 \}} f_k^3(\theta) \theta, d\theta = \int_{S^1 \cap \{ \theta, \theta < 0 \}} \theta, f_k^3(-\theta) d\theta$$

$$f_k(\theta) > f_k(-\theta).$$

Exactly two such sections are impossible for the same reason.



$$\underline{\mu(2) \geq 3}$$

$$\mu(\Delta) = 3.$$



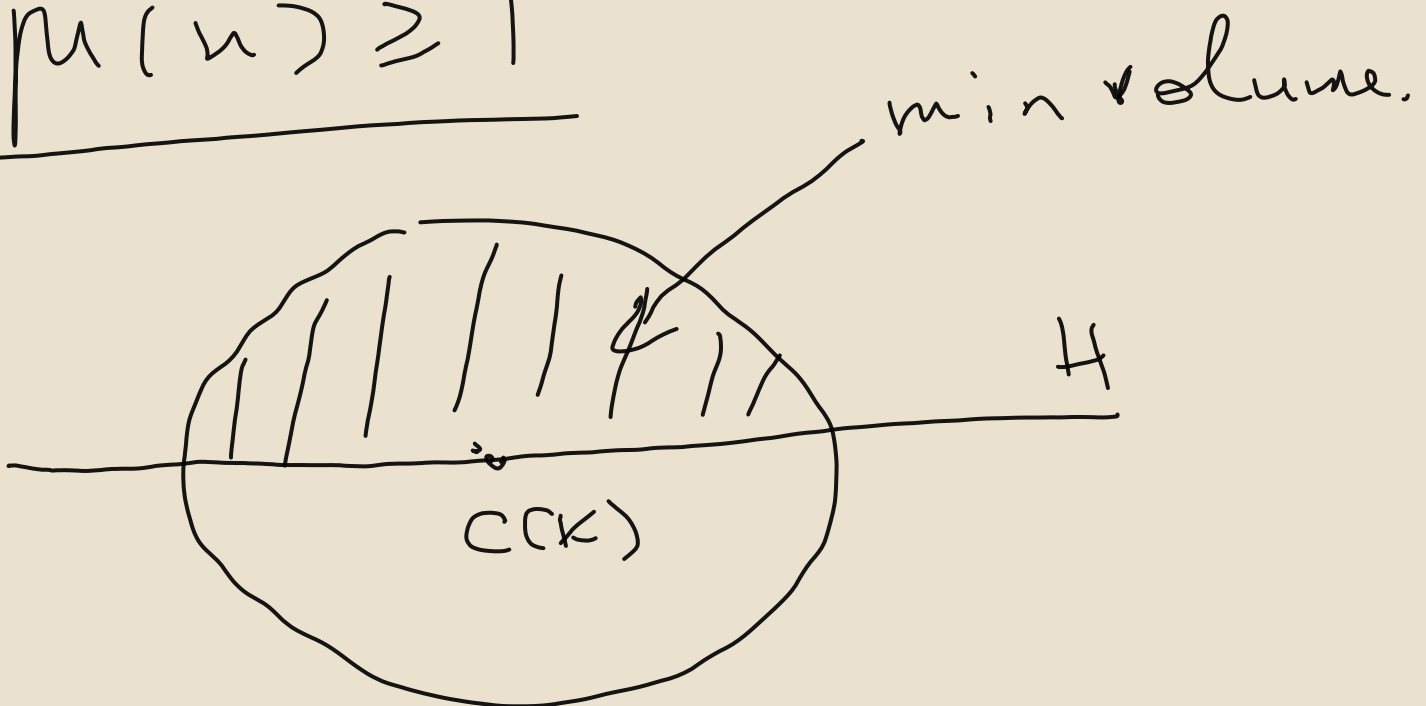
$$\underline{\mu(2) = 3}$$

We will see that

$$\mu(n) = 1$$

$$\underline{n \geq 5.}$$

$$\underline{\mu(n) \geq 1}$$



$$\text{Then } C(K \cap H) = C(K).$$

## Theorem (MT4)

$\mathcal{I}$  convex body of revolution  
about  $x_n$ -axis in  $\mathbb{R}^n$   
 $n \geq 5$

such that  $C(K) = \emptyset$

$$\langle C(K \cap \xi^\perp), e_n \rangle > 0$$

$$\forall \xi \in S^{n-1}, \xi \neq \pm e_n.$$

$$\xi^\perp = \{x \in \mathbb{R}^n; \langle x, \xi \rangle = 0\}.$$

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$\mathcal{S}'(\mathbb{R}^n)$  Schwartz space  
(test functions)

$\mathcal{S}'(\mathbb{R}^n)$  distributions  
continuous linear functionals  
on  $\mathcal{S}$

$$\varphi \in S : \text{F.T. } \hat{\varphi}(\omega) = \int_{\mathbb{R}^n} \varphi(y) e^{-i\langle x, y \rangle} dy$$

$f \in S'$  its F.T. is  $\hat{f} \in S'$

defined by

$$\langle \hat{f}, \varphi \rangle = \langle f, \hat{\varphi} \rangle$$

$$\forall \varphi \in S.$$

In particular,

if  $f$  continuous, even  
function on  $\mathbb{R}^n \setminus \{0\}$ ,

homogeneous of degree  $-n+1$ .

$$\hat{f}(\xi) = \pi \int_{S^{n-1} \cap \xi^\perp} f(\theta) d\theta$$

$$\xi \in S^{n-1}$$



# Intersection bodies

$K$  o.s. convex body in  $\mathbb{R}^n$   
is an intersection body iff

$$\widehat{f_K}(x) \geq 0$$

↑  
extended to  $\mathbb{R}^n$  with  
homogeneity -1

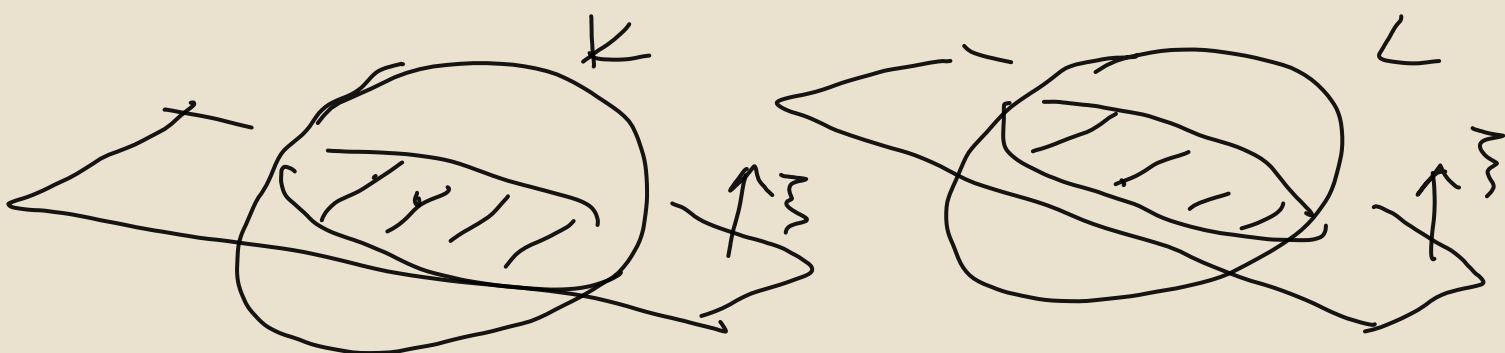
All convex o.s. bodies  
in  $\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^4$  are  
intersection bodies.

In  $\mathbb{R}^n, (n \geq 5)$  there  
are o.s. convex bodies  
that are not intersection

bodies.

BP (1956)

$K, L$  o.s. convex in  $\mathbb{R}^n$



$$\forall \xi \in S^{n-1} \quad |K \cap \xi^\perp| \leq |L \cap \xi^\perp|$$

$$\stackrel{?}{\Rightarrow} |K| \leq |L|?$$

1990's

Yes if  $n=2,$   
 $n=3,$   
 $n=4.$

No if  $n \geq 5$

# Proof of theorem

$$\int_K^n(x) = \int_M^n(x) + \varepsilon \varphi(x)$$

$M$  body of revolution about  $x_n$ -axis  
o.s. Convex body,  $C$ ,  
with positive curvature,  
that is not an  
Intersection body.

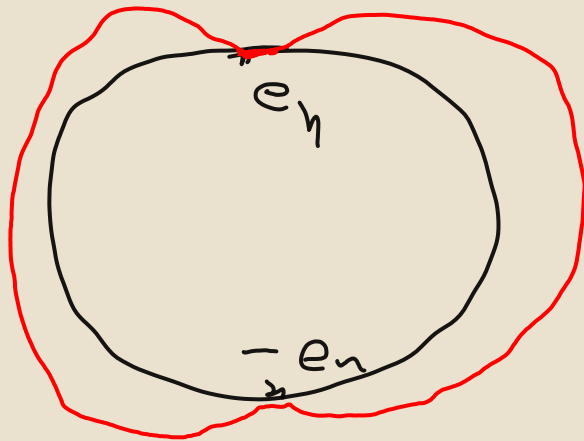
Define  $\varphi$ .

Consider  $g \in C^\infty(S^{n-1})$

- even
- invariant with respect  
to rotations about  
 $x_n$ -axis

$$\cdot g(\xi) > 0 \quad \forall \xi \in S^{n-1} \\ \xi \neq \pm e_n$$

$$\cdot g(\pm e_n) = 0.$$



Extend  $g$  to  $\mathbb{R}^n \setminus \{0\}$   
as a homog. function of  
degree  $-1$ .

Define 
$$\varphi(x) = \frac{1}{x_n} \wedge g(x)$$

$$\underline{\underline{|K \cap \xi^\perp| \langle C(K \cap \xi^\perp), e_n \rangle =}}$$

$$= \int_{K \cap \xi^\perp} x_n dx \quad x = r\theta$$

$$= \frac{1}{\kappa} \int_{S^{n-1} \cap \xi^\perp} \theta_n \rho_\kappa^n(\theta) d\theta$$

$$= \frac{\varepsilon}{\kappa} \int_{S^{n-1} \cap \xi^\perp} \theta_n \varphi(\theta) d\theta$$

$$= \frac{\varepsilon}{\pi \kappa} \left( x_n \varphi(x) \right) (\xi)$$

$$= \frac{\varepsilon}{\pi \kappa} \widehat{g(x)}(\xi) =$$

$$= \frac{(2\pi)^n \varepsilon}{\pi \kappa} g(\xi) > 0$$

$$\xi \neq \pm e_n$$

$$|k| \langle C(k), e_n \rangle$$

$$x = r\theta$$

$$= \int_K x_n dx =$$

$$= \frac{1}{n+1} \int_{S^{n-1}} \theta_n \rho_K^{n+1}(\theta) d\theta$$

$$= \frac{1}{2(n+1)} \int_{S^{n-1}} \theta_n \left( \rho_K^{n+1}(\theta) - \rho_K^{n+1}(-\theta) \right) d\theta$$

$$= \frac{1}{2(n+1)} \int_{S^{n-1}} \theta_n \left[ \left( \int_M^n (\rho_K^n(\theta) + \varepsilon \varphi(\theta)) \right)^{\frac{n+1}{n}} - \left( \int_M^n (\rho_K^n(\theta) - \varepsilon \varphi(\theta)) \right)^{\frac{n+1}{n}} \right] d\theta$$

$$= \frac{1}{2(n+1)} \int_{S^{n-1}} \theta_n \rho_K^{n+1}(\theta) \left[ \left( 1 + \frac{\varepsilon \varphi}{\rho_K^n} \right)^{\frac{n+1}{n}} - \left( 1 - \frac{\varepsilon \varphi}{\rho_K^n} \right)^{\frac{n+1}{n}} \right] d\theta$$

$$= \frac{\varepsilon}{n} \int_{S^{n-1}} \left( \theta, f_n \frac{\varphi}{f_n^n} + R_\varepsilon \right) d\theta$$

$$= \frac{\varepsilon}{n} \int_{S^{n-1}} \left( f_n \cdot \underbrace{\theta, \varphi(\theta)}_{\hat{g}} + R_\varepsilon \right) d\theta$$

$$= \frac{\varepsilon}{n} \int_{S^{n-1}} \left( f_n \cdot \hat{g} + R_\varepsilon \right) d\theta$$

$$= \frac{\varepsilon}{n} \int_{S^{n-1}} \left( \hat{f}_n \hat{g} + R_\varepsilon \right) d\theta$$

$\uparrow$  Sign-changing       $\nwarrow$  positive

$$= 0.$$



$$\mu(n) = 1 \quad n \geq 5.$$

Open question  $\mu(n)$  ?

$$\begin{array}{l} n=3 \\ n=4 \end{array}$$

Guess:  $\mu(n) = 1$   $\begin{array}{l} n=3 \\ n=4 \end{array}$



$$\delta_p = \min \frac{|K \cap H^+|}{|K|}$$

Max depth:  $\delta_{p'} = \max_{p \in K} \delta_p$

$p'$  of max depth.









